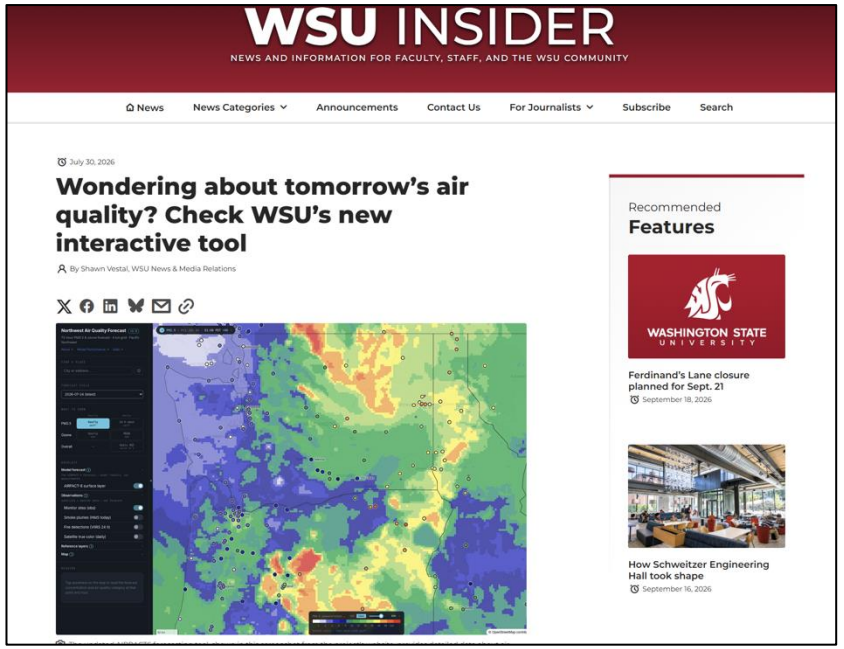
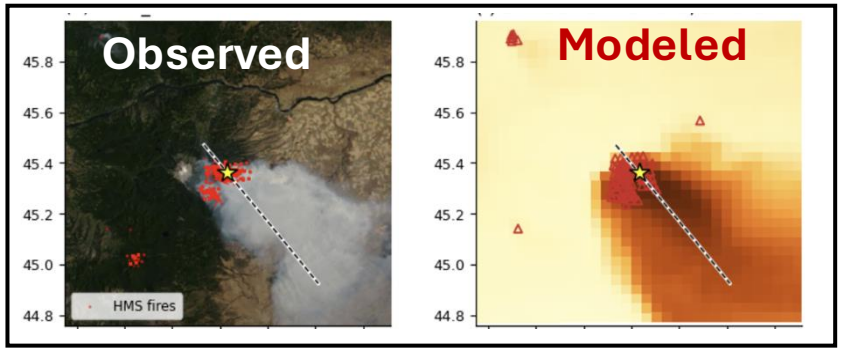
NWAIRQUEST Meeting-Oct 2026

AIRPACT6 Performance during wildfire months in 2026

Abu Ubayeda Ibnal Zarrah, AQRL at WSU





Overview

01 Operational details

02 FRP-based fire framework

03 PM2.5 performance evaluation during fire months

04 O3 performance evaluation during fire months

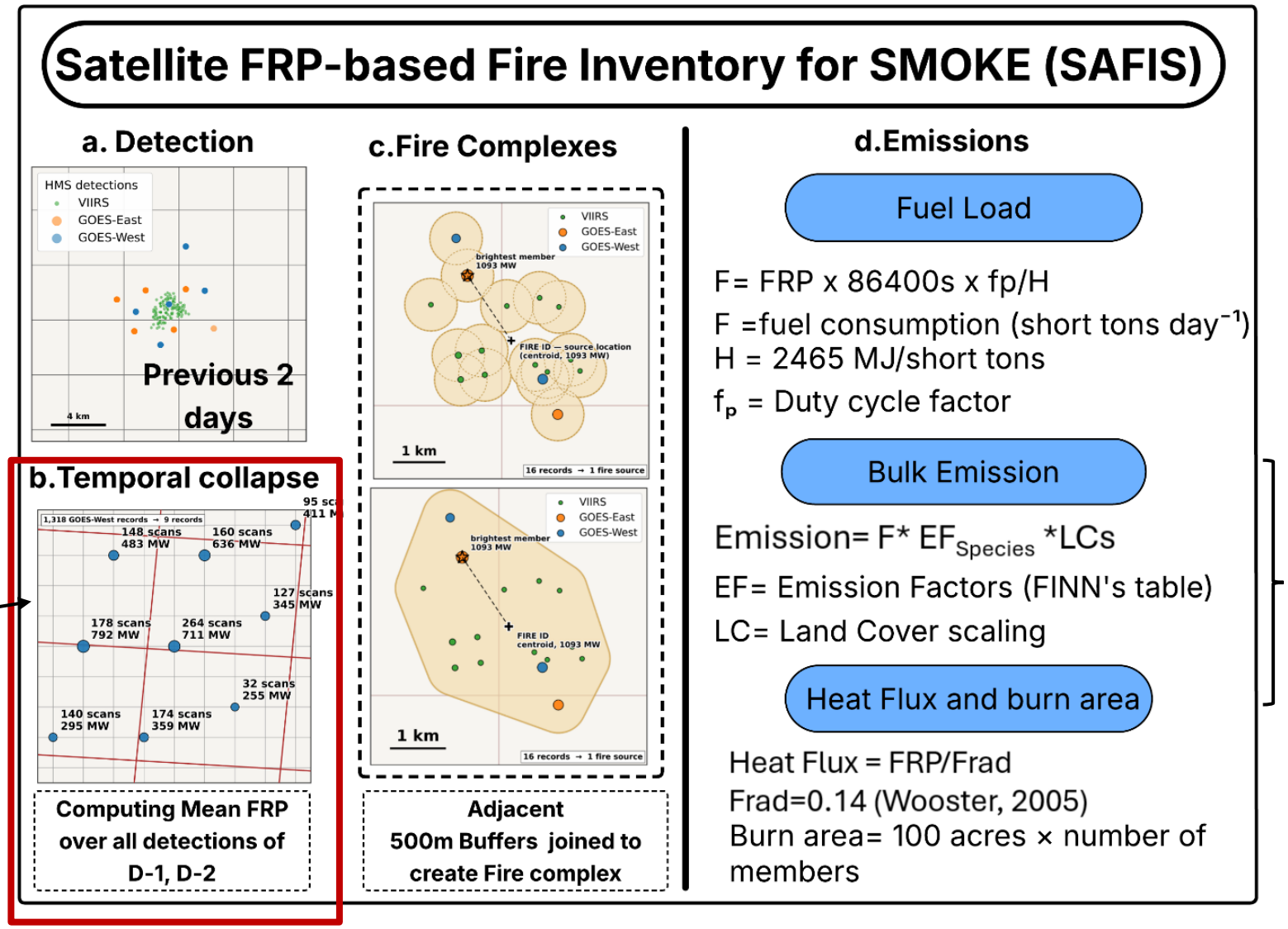
05 Summary

Current Operational System

Component	AIRPACT-5	AIRPACT-6 upgrades
CTM	CMAQ v5.0.2	CMAQ v5.5
Gas-phase mechanism	CB05	CB6r5
Aerosol module	AERO6	AE7
Emissions processor	SMOKE 3.5	SMOKE v5.2
Inventory / mobile	NEI2017 / MOVES	EPA 2022v1 / MOVES4
Biogenics	MEGAN 2.10	Inline MEGAN v3.2
Fire emissions	BlueSky (fuel loading); offline 3-D gridded	FRP-based (HMS); inline point sources; runtime Briggs plume rise

Major fire emission upgrade

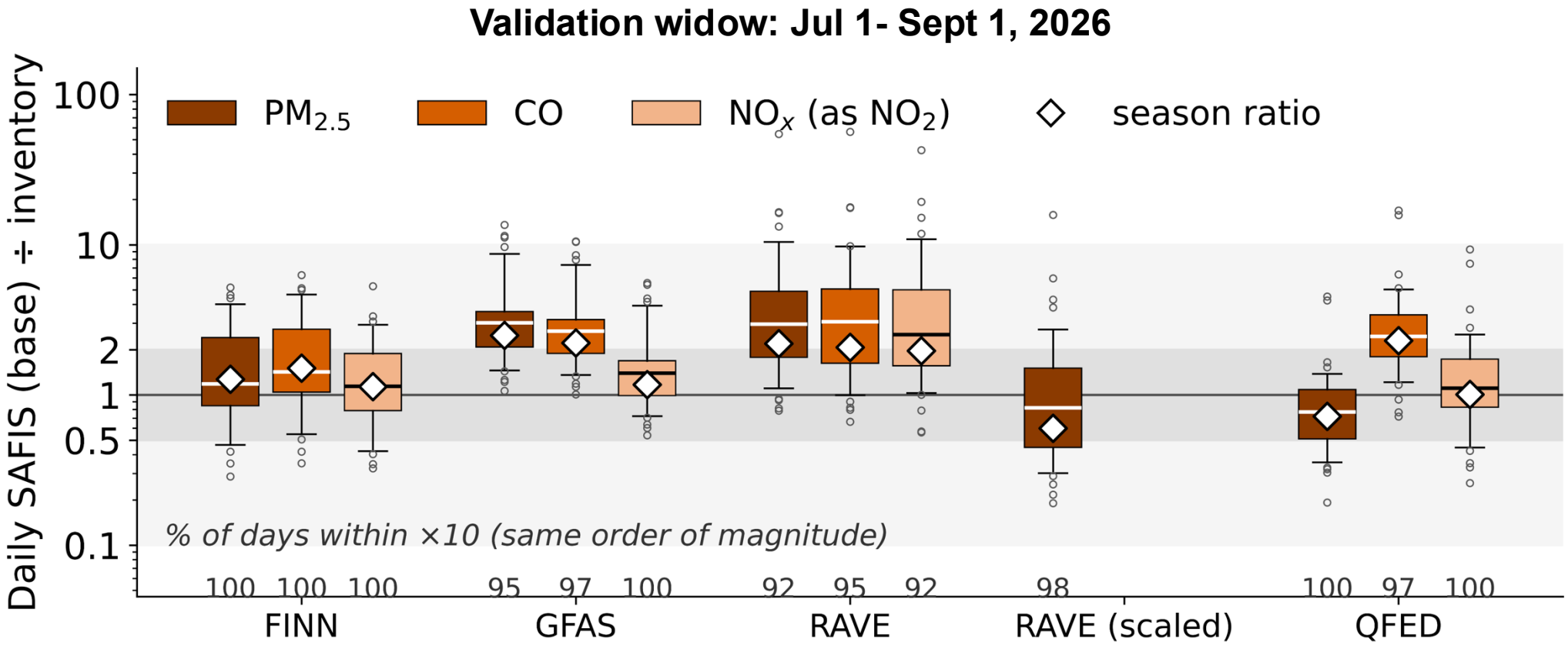
FRP based Fire Framework



Not added yet in the operational pipeline; tested in simulations shown in this talk

in the operational pipeline, EF are from AP42

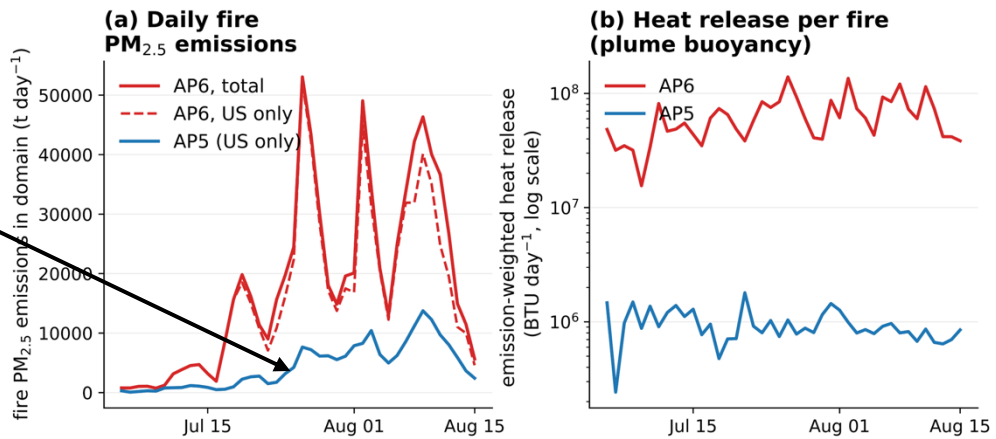
FRP based Fire Framework: Emissions

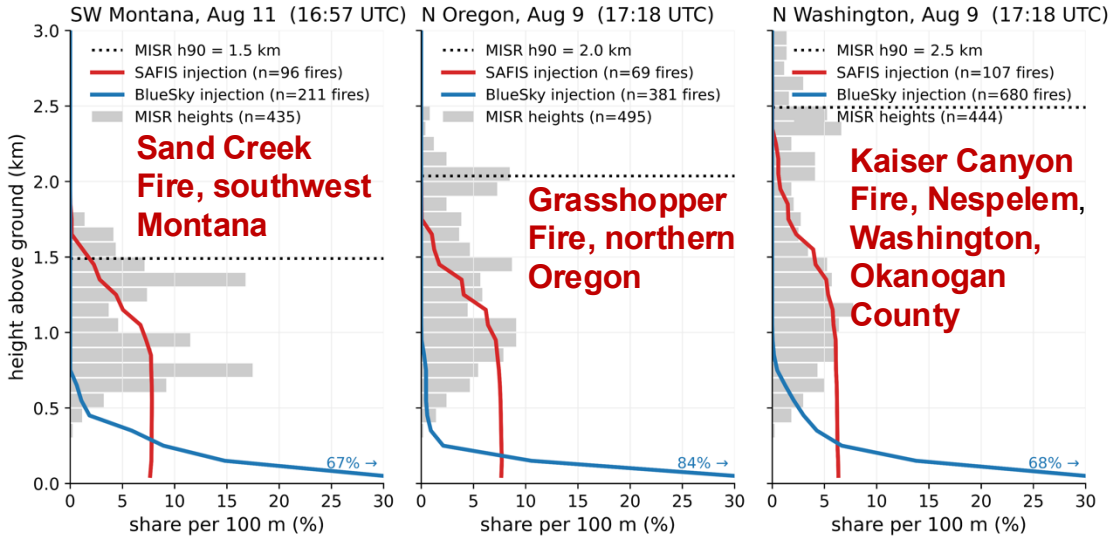


SAFIS provides reasonable emissions

FRP based Fire Framework: Physical impact

AP5 lower

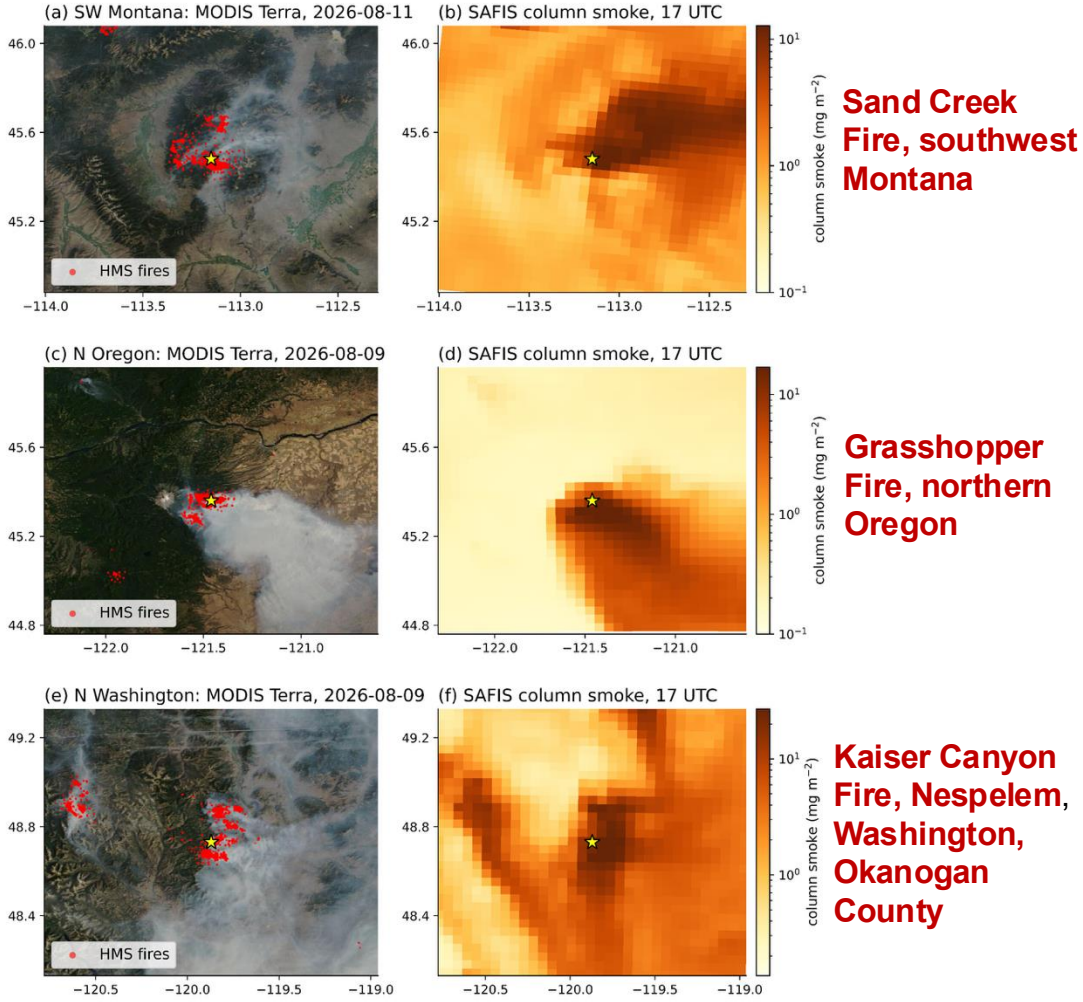




Satellite obs. from MISR

SAFIS provides more reasonable plume

Based on EC, POC, PNCOM as tracer of fresh smoke

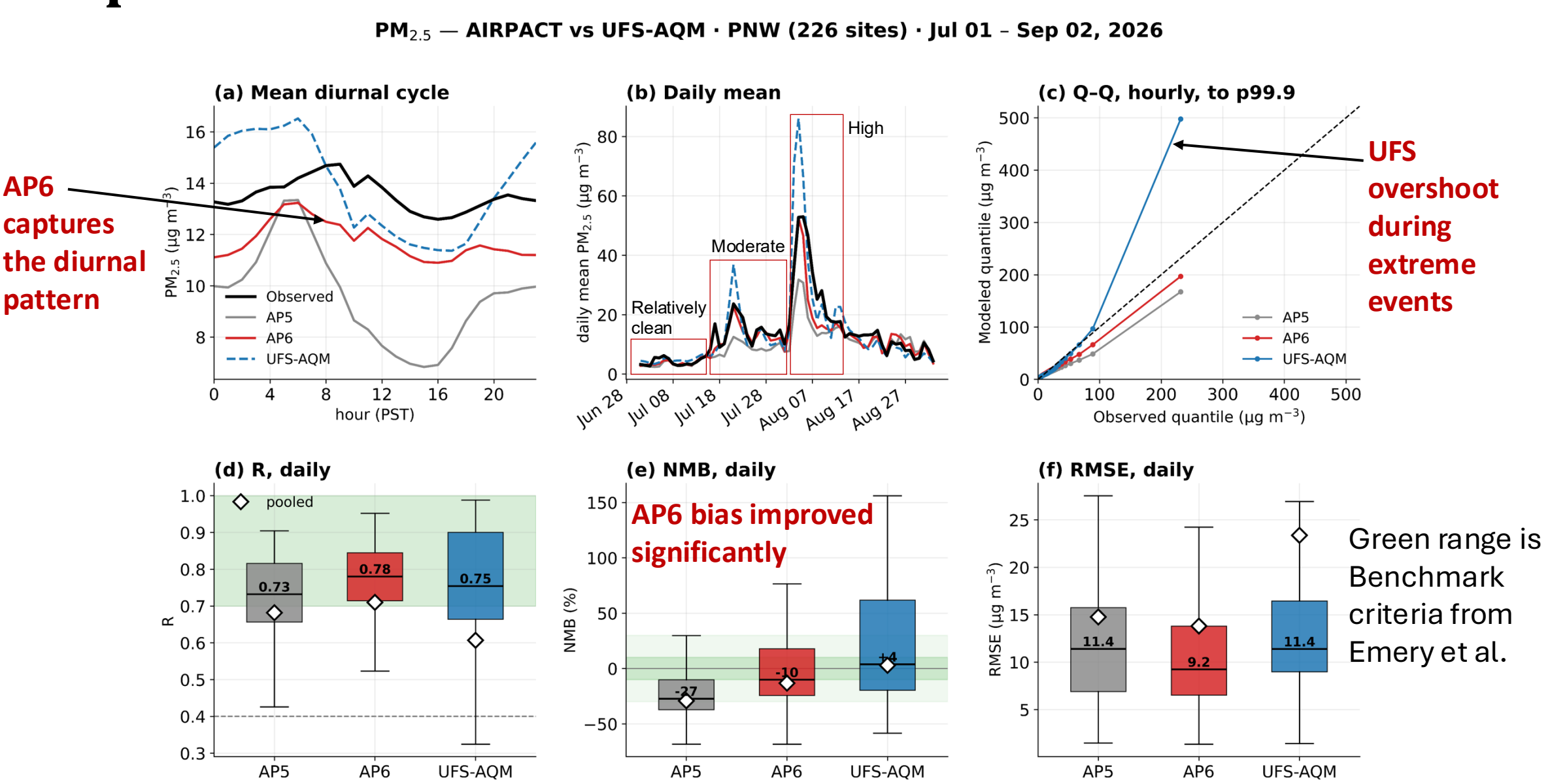


Detection and transport is reasonable

PM2.5 Performance evaluation during fire months

We used AirNow ground observations and NOAA's UFS –AQMv7

Comparison with UFS and AP5



AP6 captures the diurnal pattern

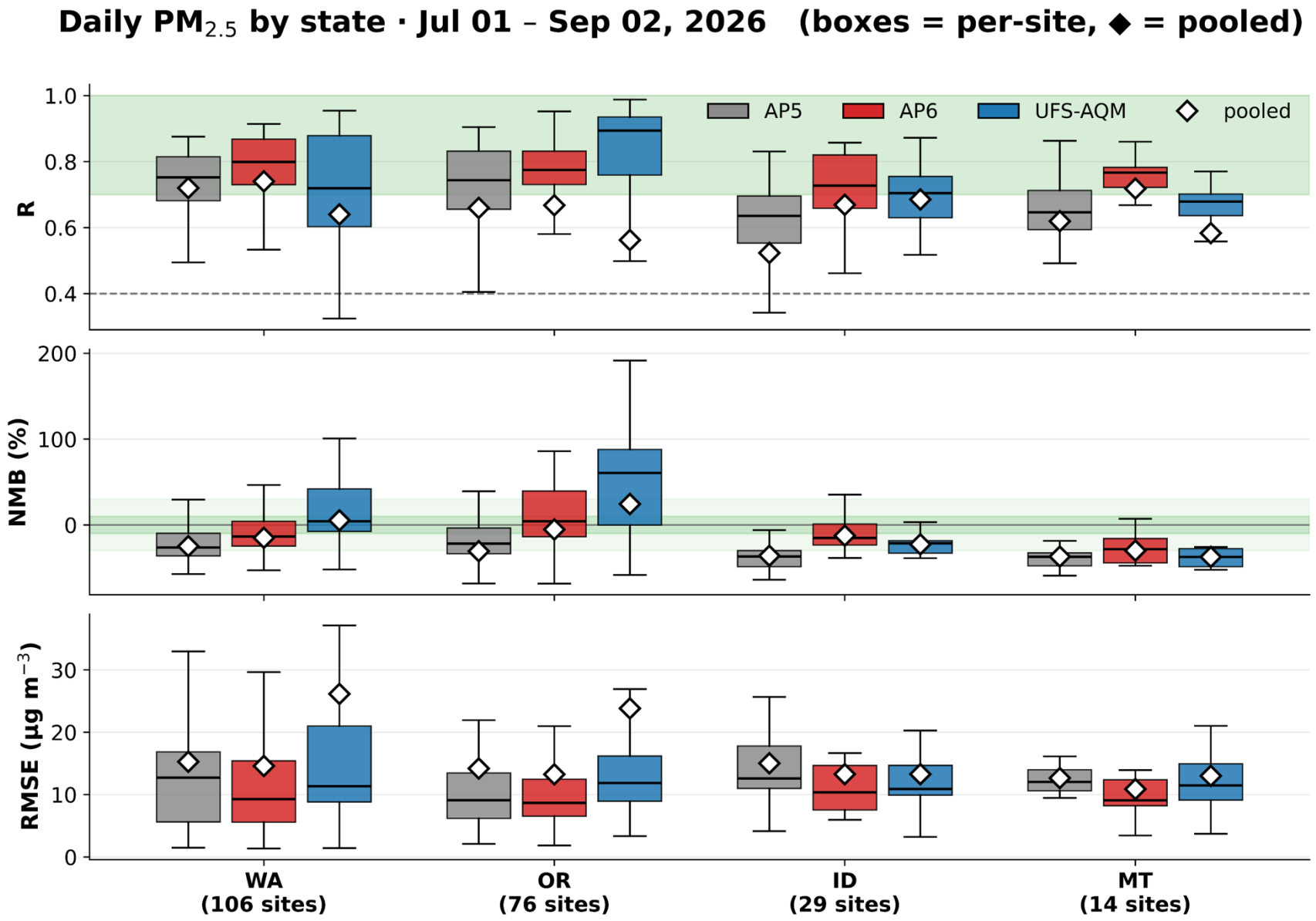
UFS overshoot during extreme events

AP6 bias improved significantly

Green range is Benchmark criteria from Emery et al.

Skill metrices improve in AP6

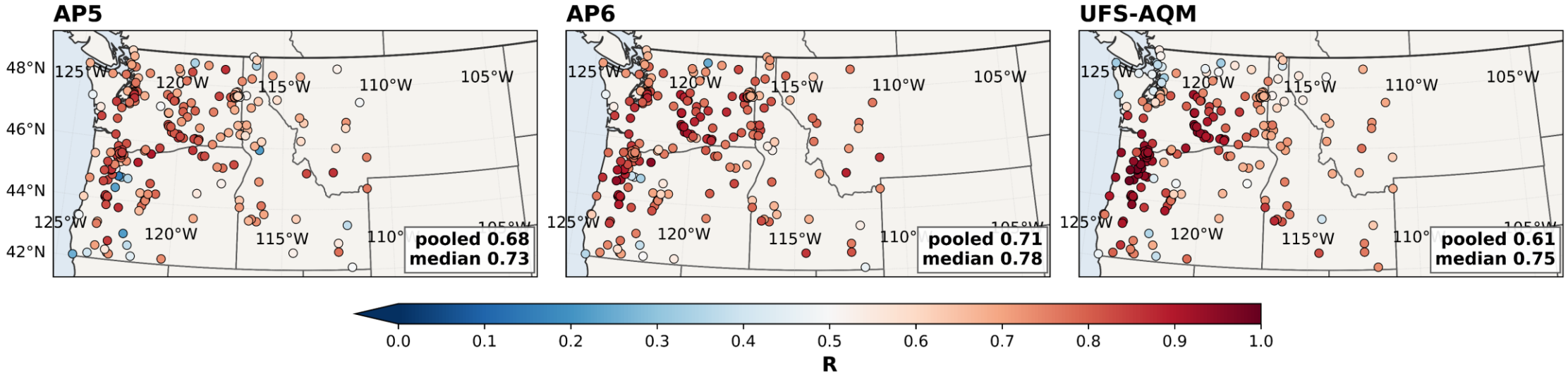
State-wise Breakdown: Skill Metrices



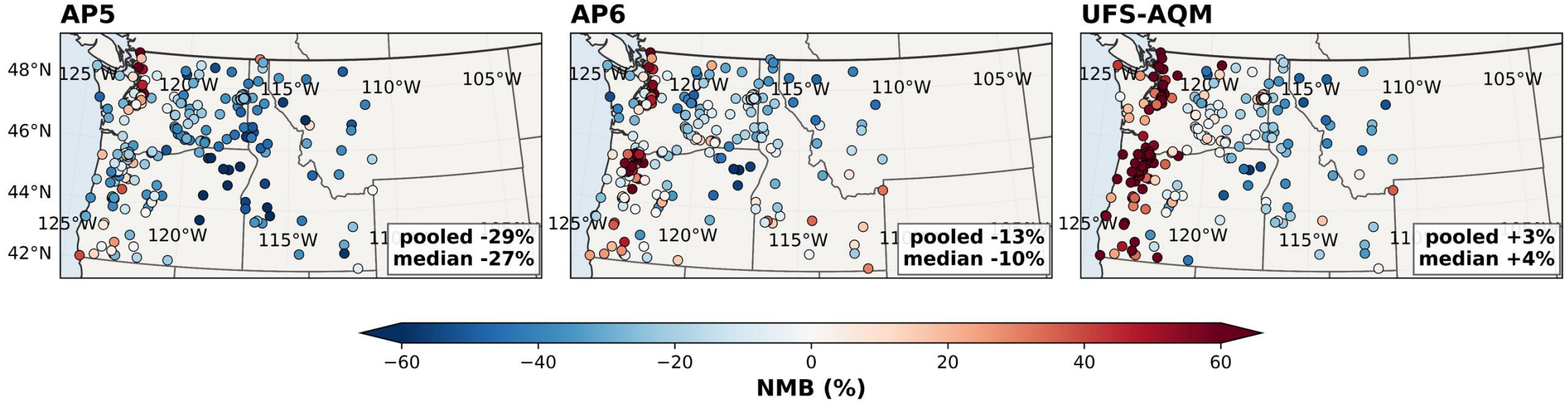
AP6 comply with the benchmark better.

AP6 residual error reduced

Spatial distribution: Skill Metrices

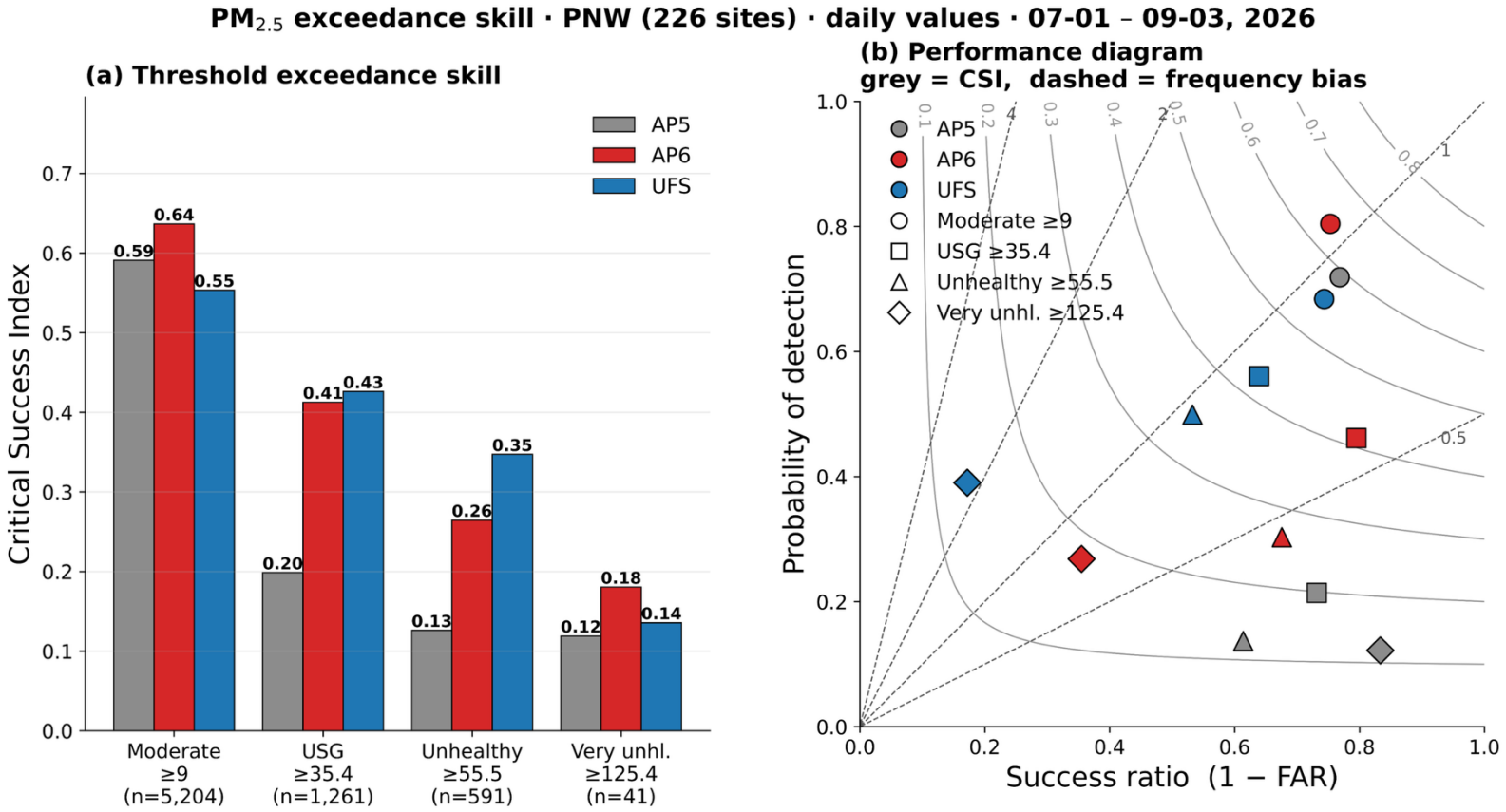


AP6 improves bias without compensating correlation



AP6 halves AP5's PM2.5 under-prediction

Exceedance Skill:



AP6 roughly doubles exceedance skill than AP5 at the smoke

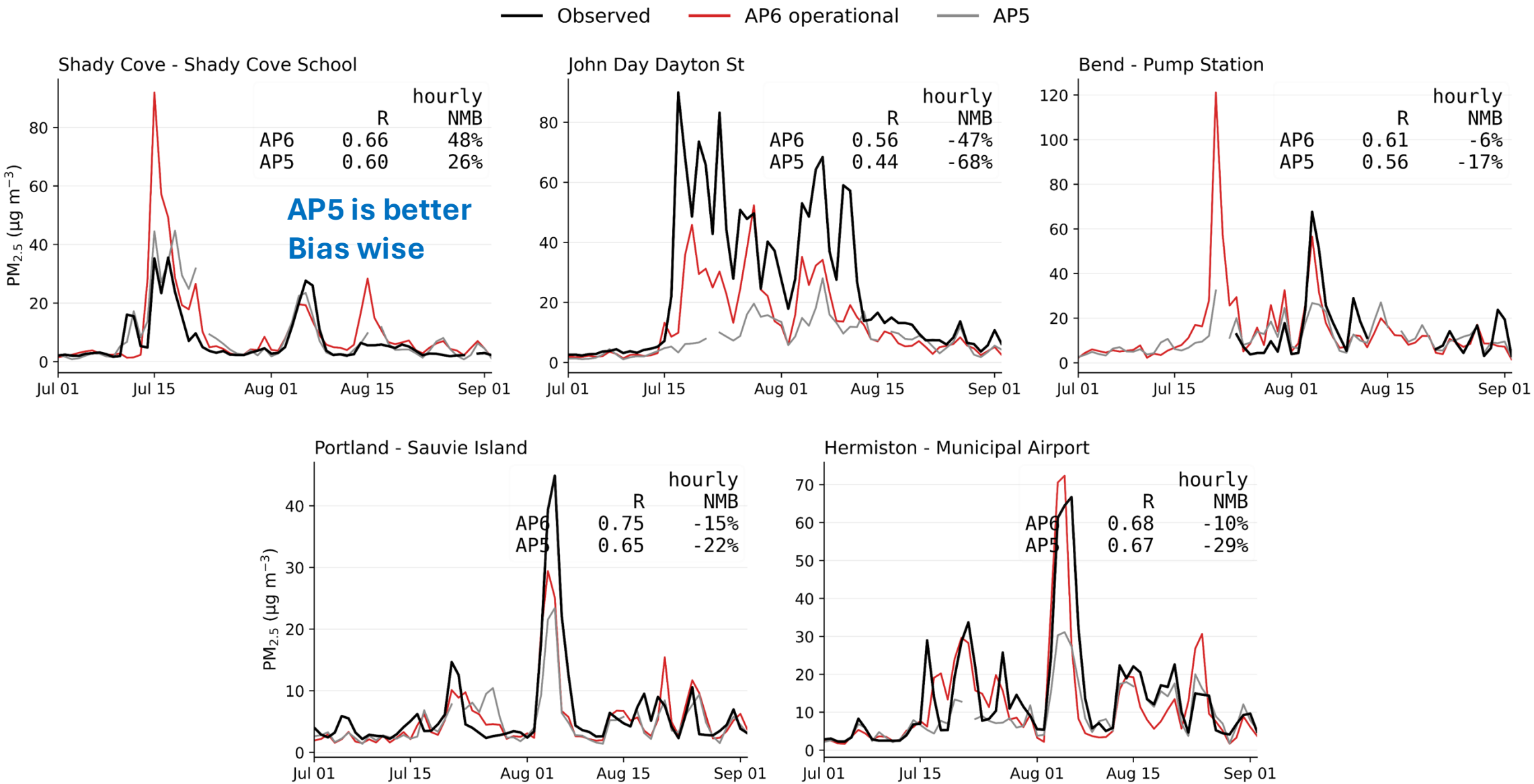
All models under-detect smoke exceedances

According to EPA guideline for forecast: https://www.airnow.gov/sites/default/files/2020-06/aq-forecasting-guidance-1016.pdf

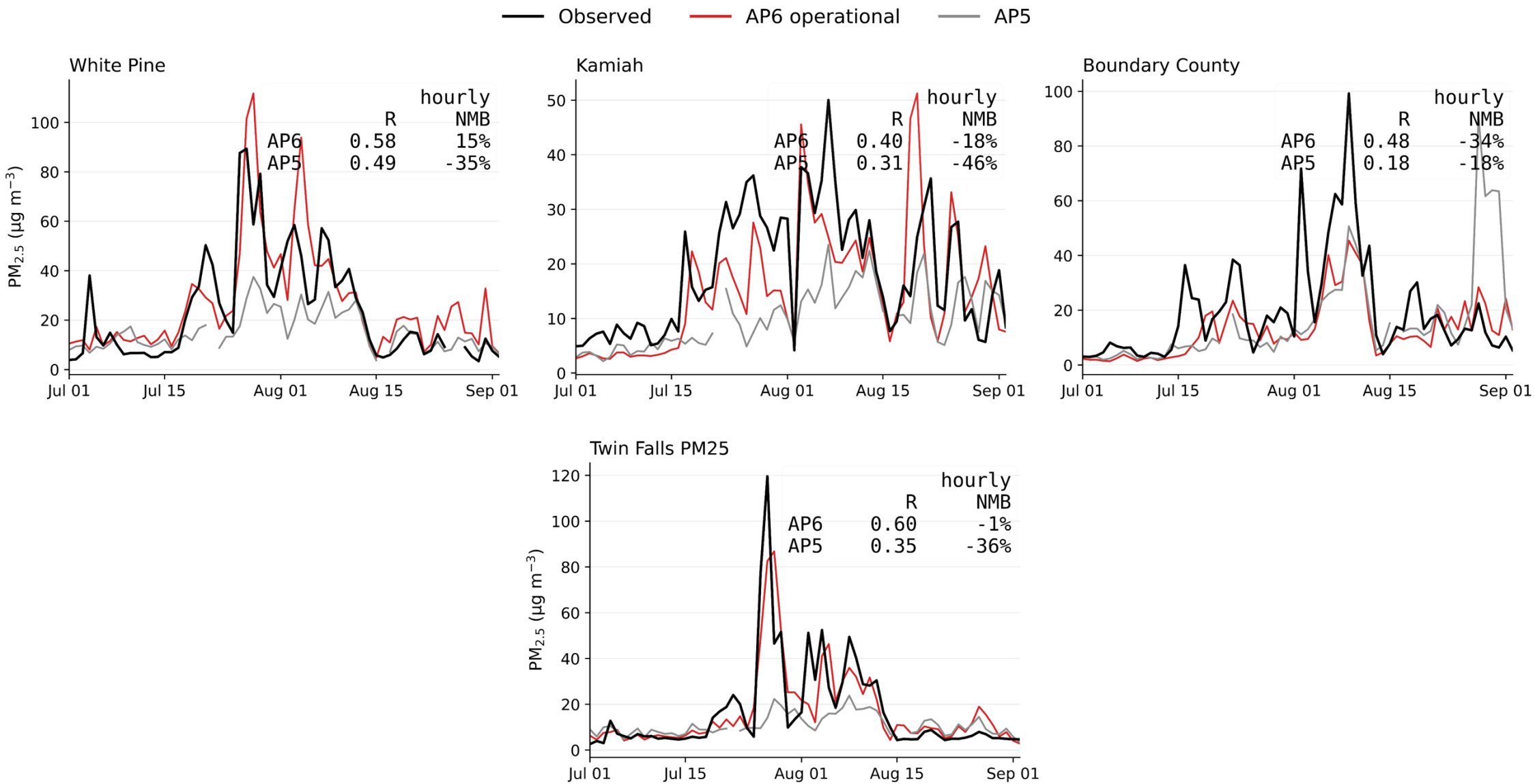
PM2.5 performance evaluation during fire months

Site-wise breakdown of PNW: AP5 and AP6
Evaluation window: July 1 to Sept 2

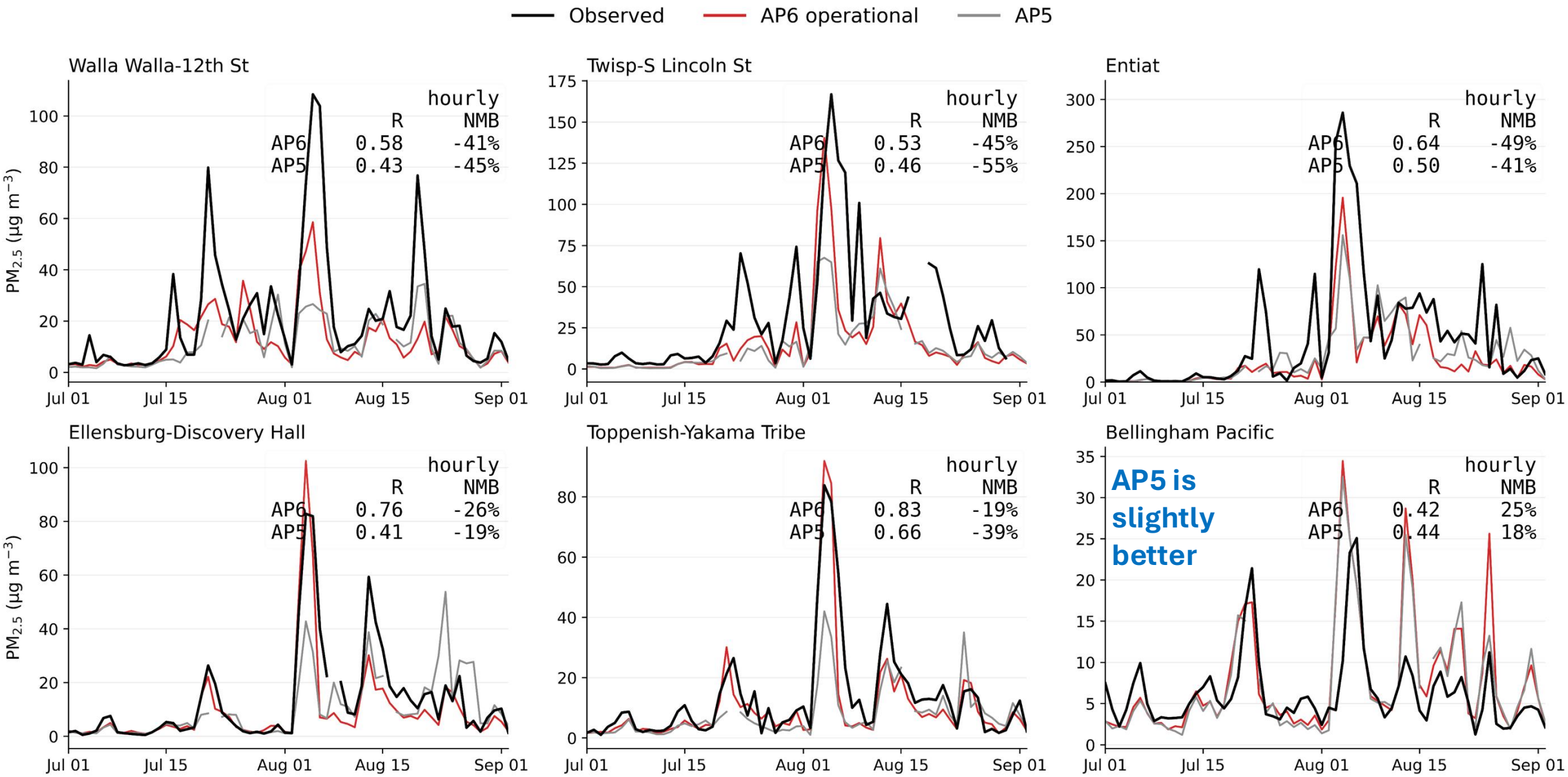
OREGON SITES



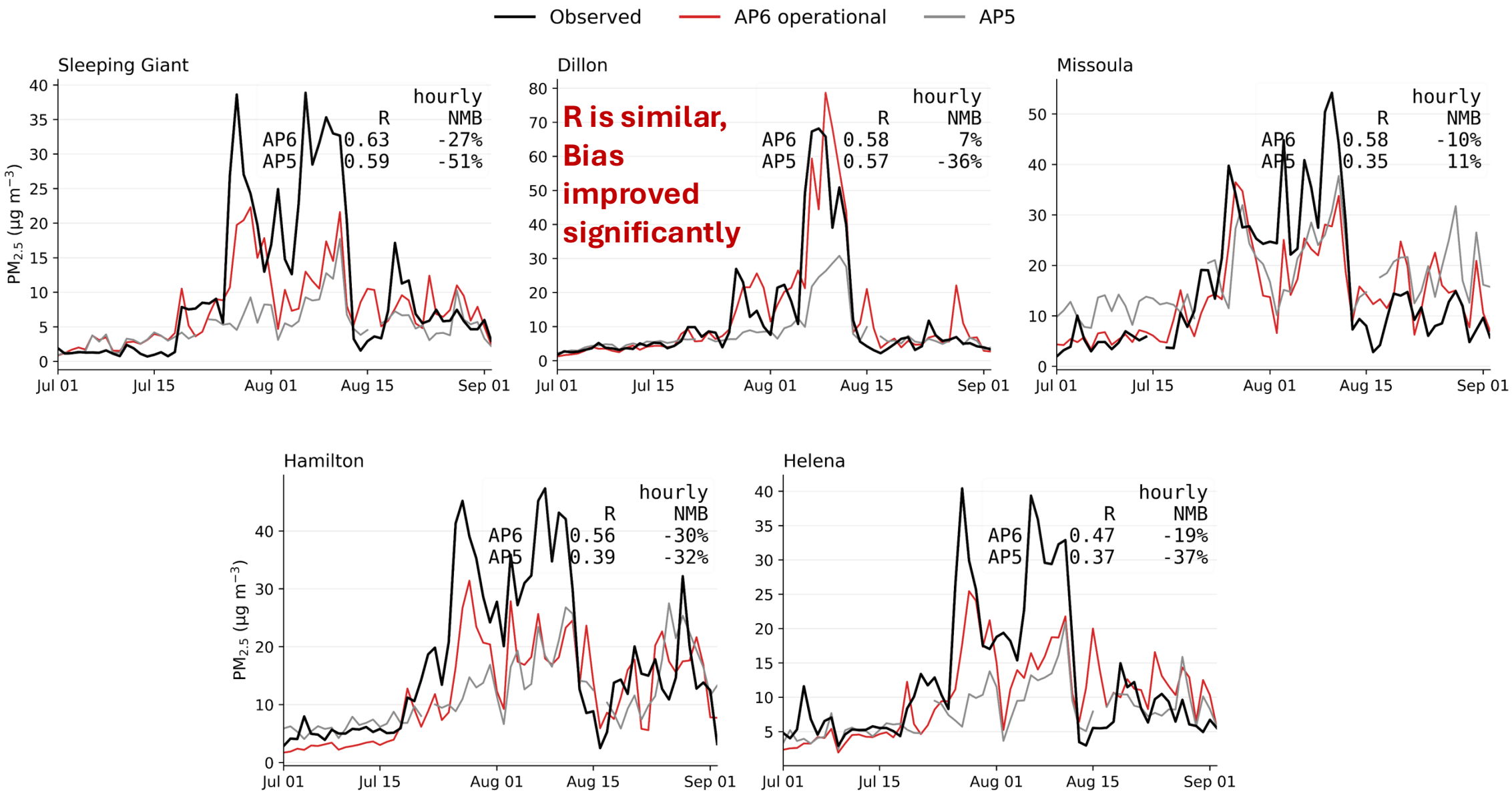
IDAHO SITES


Observed
AP6 operational
AP5
White Pine
Kamiah
Boundary County
Twin Falls PM25
hourly
R NMB
AP6 0.58 15%
AP5 0.49 -35%
AP6 0.40 -18%
AP5 0.31 -46%
AP6 0.48 -34%
AP5 0.18 -18%
AP6 0.60 -1%
AP5 0.35 -36%
PM2.5 (µg m−3)
Jul 01
Jul 15
Aug 01
Aug 15
Sep 01

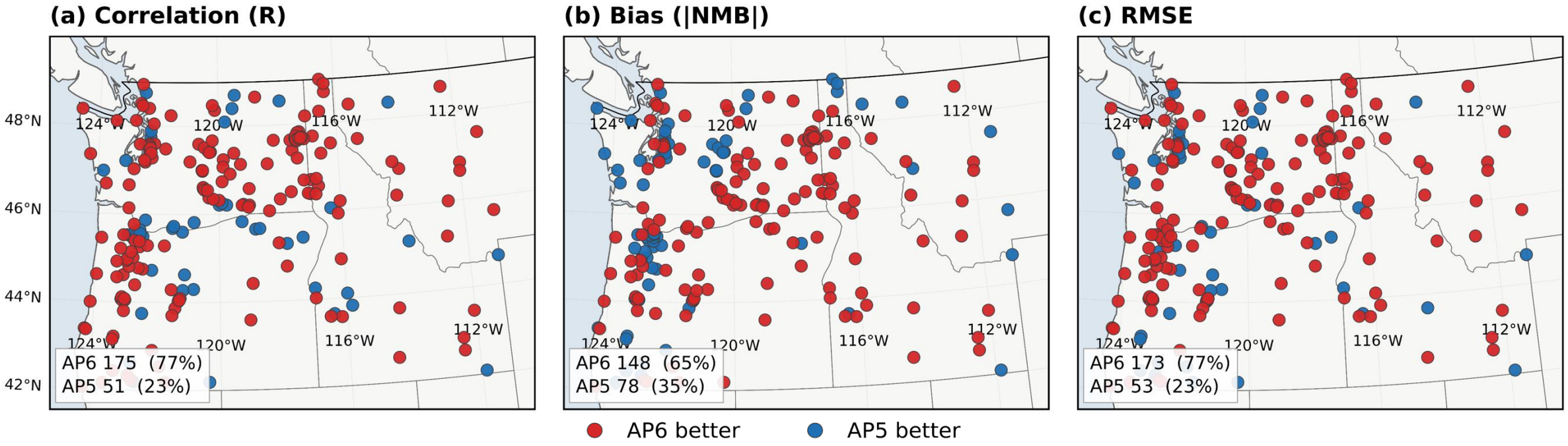
WASHINGTON SITES



MONTANA SITES



Overall Site-wise improvement



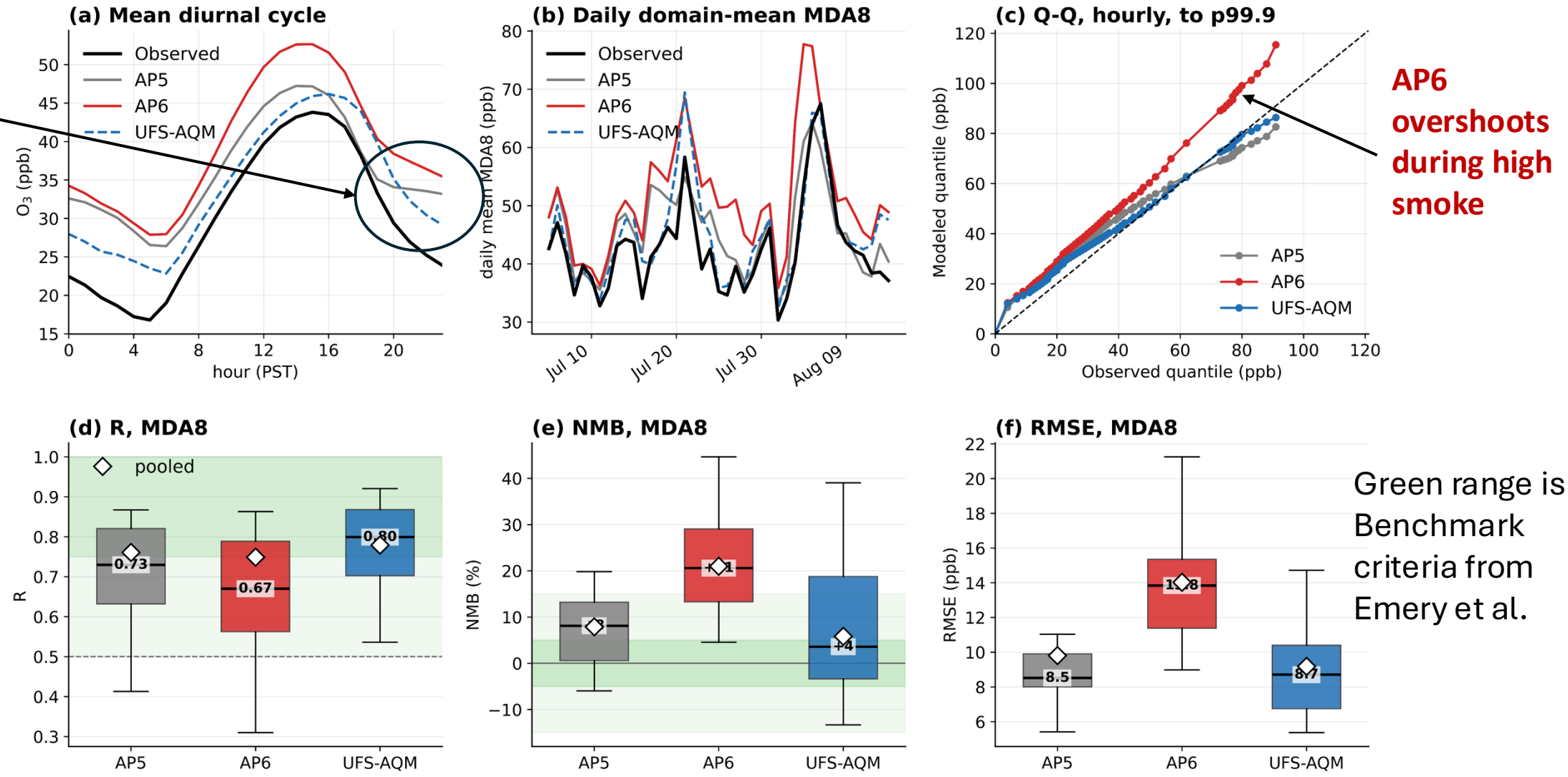
Overall AP6 performs better in most sites in 2026 fire season

Ozone performance evaluation during fire months

Site-wise breakdown of PNW: AP5 and AP6
Evaluation window: June 28 to Sept 2

Comparison with UFS and AP5

AP6 inherits same high nighttime Ozone, likely due to BLH



AP6 overshoots during high smoke

Skill metrices degrades in AP6

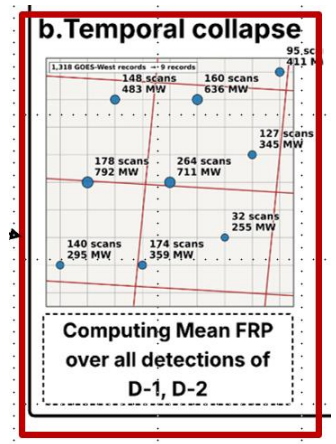
Green range is Benchmark criteria from Emery et al.

Ozone bias is higher in AP6, especially in high smoke period

Upgrades

Not yet added in operations

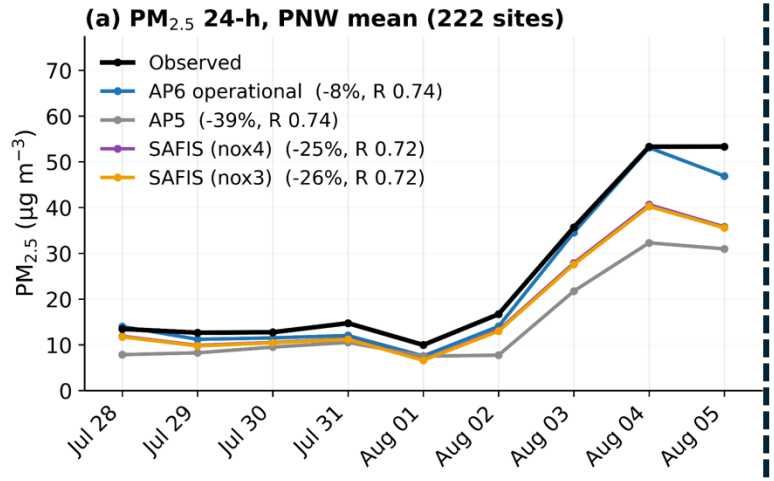
Structural Changes

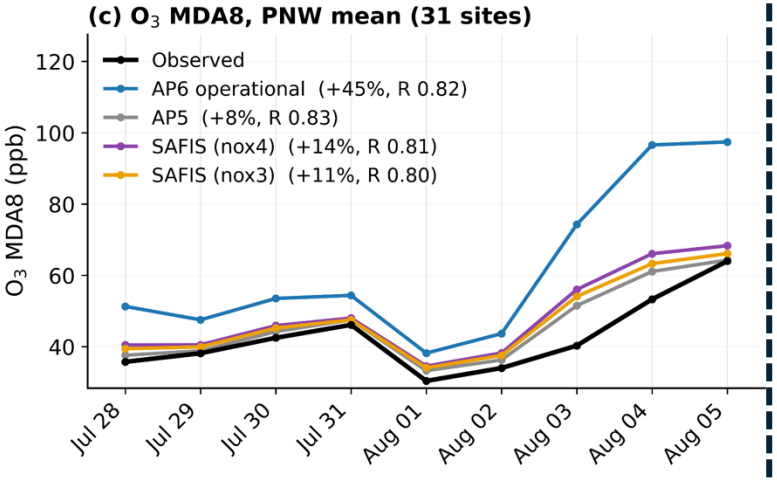


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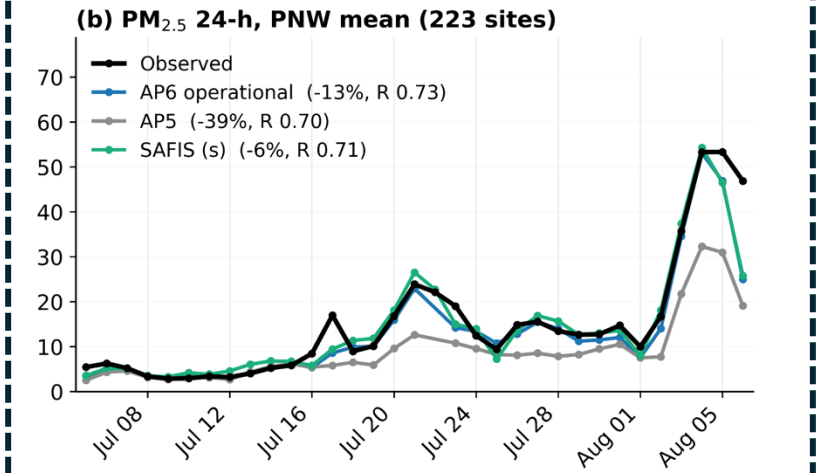
Emission factors from FINN v1

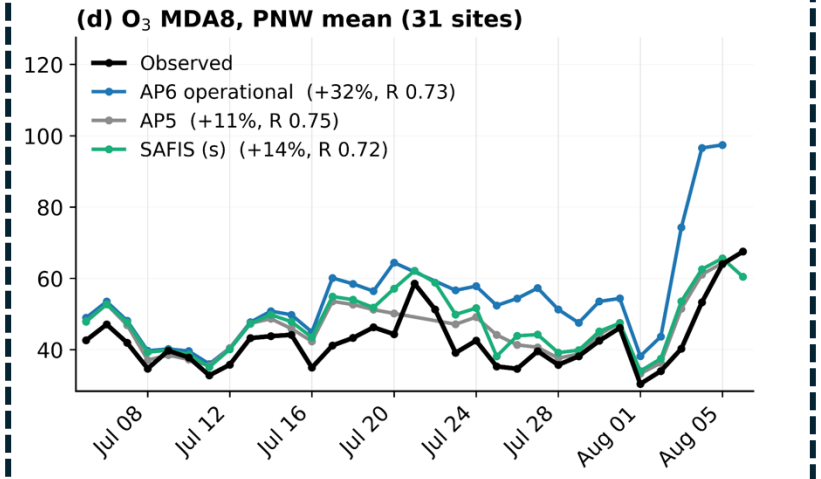
Sensitivity





Optimized





Fire Emission

PM emission x 1.5
NOx emission x0.3

Summary

01 FRP based Fire framework improves plume physics

02 FRP-based fire framework provides reasonable emissions with other inventories

03 PM2.5 performance improves, bias reduced significantly

04 AP-6 operational Ozone shows high bias.

05 Structural modification in SAFIS shows improvement for ozone without compensating PM performance

Thank you!

Questions?